Bayesian nonparametric fusion of randomised and real-world survival data for heterogeneous treatment effects

Supplementary Materials

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S.1 Proofs

S.1.1 Proof of Proposition 1

We show:

$$\mathbb{E}[\log T \mid A, X, S] \;=\; m_0(X, S) \,+\, \tau(X)\, A \,+\, (1 - S)\, A\, c(X), \tag{S1}$$

at each of the four values of (A, S): The left-hand side is the conditional mean of the observed $\log T$. The right-hand side decomposes that mean into causal quantities: a source-specific baseline $m_0(X, S)$, the treatment effect $\tau(X)$, and the confounding bias $c(X)$.

We use two elementary arguments throughout the proof. By consistency (Assumption 1), we replace the observed outcome by the relevant potential outcome on the event where treatment is fixed: $\log T = \log T(a)$ on $\{A = a\}$. By RCT unconfoundedness (Assumption 2), we have $T(a) \perp\!\!\!\perp A \mid X, S = 1$. Inside the RCT subpopulation $\{S = 1\}$, the conditional distribution of $T(a)$ given X is unchanged by further conditioning on $\{A = a\}$. The two conditional expectations coincide: $\mathbb{E}[\log T(a) \mid A = a, X, S = 1] = \mathbb{E}[\log T(a) \mid X, S = 1]$. Together, the two arguments yield, for each $a \in \{0, 1\}$:

$$\mathbb{E}[\log T \mid A = a,\; X,\; S = 1] \;=\; \mathbb{E}[\log T(a) \mid X,\; S = 1]. \tag{S2}$$

Case $A = 0,\ S \in \{0, 1\}$. The two treated-arm terms in (S1) vanish. The right-hand side reduces to $m_0(X, S)$. The left-hand side equals $m_0(X, S)$ by the definition $m_0(X, S) := \mathbb{E}[\log T \mid A = 0, X, S]$.

Case $A = 1,\ S = 1$ *(RCT)*. We apply (S2) at $a = 1$ to obtain:

$$\mathbb{E}[\log T \mid A = 1,\; X,\; S = 1] \;=\; \mathbb{E}[\log T(1) \mid X,\; S = 1]. \tag{S3}$$

We split $\log T(1) = \log T(0) + [\log T(1) - \log T(0)]$ inside the expectation:

$$\mathbb{E}[\log T(1) \mid X,\; S = 1] \;=\; \mathbb{E}[\log T(0) \mid X,\; S = 1] \,+\, \mathbb{E}[\log T(1) - \log T(0) \mid X,\; S = 1]. \tag{S4}$$

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The first term equals $m_0(X,1)$ by (S2) at $a=0$:

$$\mathbb{E}[\log T(0) \mid X,\, S=1] \;=\; \mathbb{E}[\log T \mid A=0,\, X,\, S=1] \;=\; m_0(X,1). \tag{S5}$$

The second term equals $\tau(X)$. By cross-source transportability (Assumption 4), the conditional log-treatment effect does not depend on S. The contrast on the right of (S4) therefore equals the population CATE $\tau(X) = \mathbb{E}[\log T(1) - \log T(0) \mid X]$ defined in the manuscript:

$$\mathbb{E}[\log T(1) - \log T(0) \mid X,\, S=1] \;=\; \mathbb{E}[\log T(1) - \log T(0) \mid X] \;=\; \tau(X). \tag{S6}$$

We combine the three results: $\mathbb{E}[\log T \mid A=1, X, S=1] = m_0(X,1) + \tau(X)$. This matches the right-hand side of (S1) at $(A,S) = (1,1)$. The confounding-function term $(1-S)\,A\,c(X)$ vanishes when $S=1$.

Case $A=1$, $S=0$ (RWD). We recall the definition of the confounding function from the manuscript:

$$c(x) \;=\; \mathbb{E}[\log T \mid X=x,\, A=1,\, S=0] \;-\; \mathbb{E}[\log T \mid X=x,\, A=0,\, S=0] \;-\; \tau(x). \tag{S7}$$

We rearrange:

$$\mathbb{E}[\log T \mid A=1,\, X,\, S=0] \;=\; m_0(X,0) + \tau(X) + c(X), \tag{S8}$$

which equals the right-hand side of (S1) at $(A,S) = (1,0)$.

This case uses none of the manuscript's causal-framework assumptions directly. They enter only through τ: they identify τ as the causal CATE rather than an unrelated function of X. That identification is the content of Proposition 2(i), proved below. $\square$

S.1.2 Proof of Proposition 2

We use positivity (Assumption 3 of the manuscript) throughout the proof. It guarantees that the conditional expectations below are well-defined for almost every x in the relevant support.

Part (i): identification of τ from the RCT. We evaluate (S1) at the two values of A within the RCT slice $\{S=1\}$. The confounding-function term $(1-S)\,A\,c(X)$ vanishes when $S=1$:

$$\mathbb{E}[\log T \mid A=1,\, X,\, S=1] \;=\; m_0(X,1) + \tau(X), \tag{S9}$$
$$\mathbb{E}[\log T \mid A=0,\, X,\, S=1] \;=\; m_0(X,1). \tag{S10}$$

The difference of the two conditional means cancels $m_0(X,1)$ and leaves $\tau(X)$:

$$\tau(X) \;=\; \mathbb{E}[\log T \mid A=1,\, X,\, S=1] \;-\; \mathbb{E}[\log T \mid A=0,\, X,\, S=1]. \tag{S11}$$

Both conditional means on the right are functionals of the joint distribution of the observed outcomes $(\log T, A, X) \mid S=1$. The RCT observes this distribution. This identifies τ from the RCT alone.

The identification rests on two assumptions. By RCT unconfoundedness (Assumption 2 of the manuscript), the observed RCT contrast at $X=x$ equals the conditional treatment effect within the RCT population. By cross-source transportability (Assumption 4 of the manuscript), this RCT-population effect coincides with the population CATE. Neither the RWD data nor the confounding function c enters (S11).

Part (ii): identification of c from the RWD, given τ. We now evaluate (S1) at the two values of A within the RWD slice $\{S=0\}$:

$$\mathbb{E}[\log T \mid A=1,\, X,\, S=0] \;=\; m_0(X,0) + \tau(X) + c(X), \tag{S12}$$
$$\mathbb{E}[\log T \mid A=0,\, X,\, S=0] \;=\; m_0(X,0). \tag{S13}$$

The difference cancels $m_0(X, 0)$. We rearrange the result for c:

$$c(X) \;=\; \mathbb{E}[\log T \mid A=1,\, X,\, S=0] \;-\; \mathbb{E}[\log T \mid A=0,\, X,\, S=0] \;-\; \tau(X). \tag{S14}$$

The right-hand side is the defining equation (S7) of the confounding function. The first two terms are functionals of the RWD data distribution. If τ is already identified, then (S14) identifies c. Part (i) supplies such a τ from the RCT.

Part (iii): the RWD alone does not identify τ and c separately. We show that distinct parameter pairs (τ, c) produce the same RWD distribution. The decomposition (S1) restricted to $\{S = 0\}$ yields two equations in three unknown functions $m_0(\cdot, 0)$, τ, and c:

$$\mathbb{E}[\log T \mid A=0,\, X,\, S=0] \;=\; m_0(X,0), \tag{S15}$$

$$\mathbb{E}[\log T \mid A=1,\, X,\, S=0] \;=\; m_0(X,0) + \tau(X) + c(X). \tag{S16}$$

The first equation identifies $m_0(\cdot, 0)$. The second equation identifies the composite $\tau + c$ but cannot separate its two components.

We make the ambiguity explicit. Fix any function $g : \mathcal{X} \to \mathbb{R}$ and set $\tilde{\tau}(x) := \tau(x) + g(x)$ and $\tilde{c}(x) := c(x) - g(x)$. The sum is preserved: $\tilde{\tau}(x) + \tilde{c}(x) = \tau(x) + c(x)$ for every x. Substitution of $(\tilde{\tau}, \tilde{c})$ for (τ, c) into (S16) reproduces the same RWD conditional means. The two parameter pairs are observationally equivalent on the RWD slice.

Two natural choices of g illustrate the ambiguity. Take $g \equiv -\tau$. Then $\tilde{\tau} \equiv 0$ and $\tilde{c} \equiv \tau + c$: the entire RWD contrast is attributed to confounding bias, with no causal effect. Take $g \equiv c$. Then $\tilde{\tau} \equiv \tau + c$ and $\tilde{c} \equiv 0$: the entire RWD contrast is attributed to a causal effect, under the false premise that the RWD is unconfounded. The two interpretations are indistinguishable from the RWD data alone. The non-identification is structural rather than statistical. Larger RWD samples cannot resolve it. External information that identifies one of the two functions is required. $\square$

S.2 The hierarchical m_0 prior as a meta-analytic-predictive prior

The meta-analytic-predictive (MAP) prior of Neuenschwander et al. (2010) is a hierarchical Bayesian construction for borrowing strength between studies. We argue that the main-text specification (11) of $m_0(X, S)$ is a function-valued, heterogeneous MAP prior. The symmetric form of this prior is the canonical MAP construction. The main text uses its two-study instance, anchored at the RCT, which we derive below.

The MAP prior assumes a Gaussian hierarchical structure across K studies with study-specific parameters $\theta_1, \ldots, \theta_K$:

$$\theta_k \mid \theta^*, \sigma_d^2 \sim N(\theta^*, \sigma_d^2), \qquad k = 1, \ldots, K, \tag{S17}$$

$$\theta^* \sim p(\theta^*), \tag{S18}$$

$$\sigma_d \sim p(\sigma_d). \tag{S19}$$

The shared mean θ^* is the borrowing target. The heterogeneity variance σ_d^2 controls how strongly each θ_k is shrunk towards θ^*. Small σ_d enforces near-equality of the θ_k, i.e. strong borrowing. Large σ_d allows the θ_k to drift apart, i.e. weak borrowing. The hierarchical structure lets the data determine the degree of borrowing through the posterior of σ_d.

We restrict the MAP prior to two studies before extending it to functions. Write $\theta_s = \theta^* + d_s$ with $d_s \sim N(0, \sigma_d^2)$ for the RWD ($s = 0$) and the RCT ($s = 1$). The model carries three latent quantities, θ^*, d_0, and d_1, but the two studies determine only the two parameters θ_0 and θ_1. The overall level of θ^* is therefore not separately identified from the study-specific deviations. We resolve this by anchoring at the RCT. Take a vague prior on the shared mean, $\theta^* \sim N(0, \kappa^2)$ with $\kappa^2 \to \infty$. The pair (θ_0, θ_1) is then jointly Gaussian with $\mathrm{Var}(\theta_s) = \kappa^2 + \sigma_d^2$ and $\mathrm{Cov}(\theta_0, \theta_1) = \kappa^2$. The contrast satisfies $\theta_0 - \theta_1 = d_0 - d_1 \sim N(0,\, 2\sigma_d^2)$. We condition on the RCT parameter θ_1 and let $\kappa^2 \to \infty$. The regression coefficient then tends to one and the conditional variance tends to $2\sigma_d^2$:

$$\theta_0 \mid \theta_1 \sim N\Big(\theta_1,\; 2\sigma_d^2\Big). \tag{S20}$$

The shared level θ^* has dropped out, and the RCT parameter θ_1 now serves as the borrowing target. This anchoring at the RCT is a reduction forced by having two studies and a non-informative level, not an extra assumption. Equation (S20) also reads as an informative prior that centres the RWD parameter on the RCT parameter, with a heterogeneity scale that controls the borrowing (Schmidli et al., 2014).

We now extend the scalar reduction to functions. The study-specific parameter becomes the prognostic function evaluated at source s, $\theta_s(X) := m_0(X, s)$, and the Gaussian deviation becomes a mean-zero BART deviation. The conditional form (S20) then gives the main-text specification (11):

$$m_0(X, S) = m_0^{\mathrm{sh}}(X) + (1 - S) \cdot d(X). \tag{S21}$$

We set the shared function equal to the RCT prognosis, $m_0^{\mathrm{sh}} := m_0(\cdot, 1)$, and write the single RWD deviation as d, with $d \sim \mathrm{BART}(\sigma_{h,d})$ and mean zero. Here $\mathrm{BART}(\sigma_h)$ denotes a BART prior indexed by its leaf scale σ_h, suppressing the tree-structure arguments of the full notation $\mathrm{BART}(J, k, \alpha, \beta)$. The deviation d carries the between-source heterogeneity, and the prior is heterogeneous because d varies with X. By (S20) the scale of d is the symmetric deviation scale inflated by $\sqrt{2}$, which we absorb into the calibration of k_d below. In the functional model we do not take the vague-mean limit literally. We instead assign m_0^{sh} its own $\mathrm{BART}(200, \ldots)$ prior as a regularised borrowing target, and use the limit only to justify the anchoring. For more than two sources, or when neither source is a natural reference, we keep a separate mean-zero deviation d_s for each source, which retains the same identifiability of τ and c.

We treat the borrowing scale as a calibrated hyperparameter rather than sampling it. The leaf scale of the deviation forest is set by the Chipman–George–McCulloch rule (Chipman et al., 2010):

$$\sigma_{h,d} \;=\; \frac{k_d}{2\sqrt{J_d}}, \tag{S22}$$

where J_d is the number of trees in the deviation forest and $k_d > 0$ is the tuning constant of the Chipman calibration. The marginal standard deviation of $d(X)$ is then $\sqrt{J_d}\,\sigma_{h,d} = k_d/2$, independent of J_d. So k_d is the borrowing-strength dial. Small k_d shrinks d towards zero and induces strong borrowing of the RWD prognostic towards the RCT prognostic. Large k_d relaxes the prior and lets d depart further from zero. We take $k_d = 1$ by default, in line with the uniform Chipman calibration of the main text, and the $\sqrt{2}$ factor from the anchored reduction (S20) is absorbed into this choice. A fully Bayesian alternative is to put a hyperprior on the borrowing scale as in the standard and robust MAP (Neuenschwander et al., 2010; Schmidli et al., 2014); this trades the simplicity of the BART convention for additional data-driven adaptation of the borrowing strength.

The standard MAP (S19) assumes Gaussian deviations, which is restrictive when one or more studies are outliers. Hupf et al. (2021) extend MAP to flexible deviations through a Dirichlet-process-based prior on the heterogeneity distribution. The BART deviation in our prior is flexible in a similar spirit, though along a different axis. Hupf et al. (2021) make the distribution of deviations across studies flexible, which guards against an outlying study. Our deviation instead lets the shape of d vary flexibly with the covariates X, while retaining the borrowing target m_0^{sh}.

The construction justifies the two-forest decomposition of the baseline prognosis. A single BART on $m_0(X, S)$ with S as a covariate cannot encode the preference for borrowing, and two independent baselines would imply a needlessly diffuse prior on their difference. The shared-plus-deviation form instead places the prior directly on the between-source difference and tunes the borrowing through the deviation scale.

S.3 Posterior computation

We sample the posterior by a blocked Gibbs sampler. We recap the model, summarise one iteration of the outer Gibbs sampler, and derive the update for each parameter block in turn. We close with the calibration of the hyperparameters specific to the error model.

S.3.1 Model recap and augmented representation

The Bayesian fusion forest of the main text decomposes the log survival time as:

$$\log T_i \;=\; m_0^{\mathrm{sh}}(x_i) + (1-s_i)\,d(x_i) + a_i\,\tau(x_i) + (1-s_i)\,a_i\,c(x_i) + \varepsilon_i, \tag{S23}$$

with $s_i \in \{0,1\}$ the source indicator ($s_i = 1$ for RCT, $s_i = 0$ for RWD), $a_i \in \{0,1\}$ the treatment indicator, and $x_i \in \mathbb{R}^p$ the covariate vector. The four regression functions $f \in \{\mathrm{sh}, d, \tau, c\}$ carry independent BART priors. The errors satisfy $\varepsilon_i \mid S_i = s \sim F_s$, with F_s a centred location mixture of Gaussians:

$$\varepsilon_i \mid S_i = s,\, G_s,\, \sigma_s \;\sim\; \int \frac{1}{\sigma_s}\,\phi\!\left(\frac{w-\theta}{\sigma_s}\right) dG_s(\theta), \qquad G_s = \sum_k \pi_{sk}\,\delta_{\theta_k^*-\mu_s}, \tag{S24}$$

with shared atoms θ_k^*, source-specific weights $\boldsymbol{\pi}_s = (\pi_{sk})$, and source-specific shifts $\mu_s = \sum_k \pi_{sk}\theta_k^*$ that enforce $\mathbb{E}[\varepsilon_i \mid S_i = s] = 0$. The mixing measures are tied across sources through a hierarchical Dirichlet process with concentrations γ (top-level) and M_0, M_1 (source-specific), and per-source residual scales σ_s.

We observe each subject as a triple (L_i, R_i, δ_i) with $T_i \in [L_i, R_i]$ and $\delta_i \in \{0,1,2\}$ encoding the censoring type. The value $\delta_i = 1$ corresponds to exact observation ($L_i = R_i = T_i$). The value $\delta_i = 0$ corresponds to right censoring ($L_i = C_i,\ R_i = \infty$). The value $\delta_i = 2$ corresponds to interval censoring ($0 \le L_i < R_i < \infty$). The sampler treats $\log T_i$ as a latent quantity for every subject with $\delta_i \neq 1$, and augments it at the end of each iteration.

For each observation we define the partial residual on the augmented log-time scale:

$$R_i \;:=\; \log T_i - m_0^{\mathrm{sh}}(x_i) - (1-s_i)\,d(x_i) - a_i\,\tau(x_i) - a_i(1-s_i)\,c(x_i). \tag{S25}$$

The residual collects the contribution of the error ε_i at the current draw of the regression functions.

For computation we truncate the top-level stick-breaking at a finite level K (Sethuraman, 1994; Ishwaran and James, 2001). The top-level weights $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_K)$ are constructed by $\beta_k = u_k \prod_{l<k}(1-u_l)$ with $u_k \overset{\mathrm{iid}}{\sim} \mathrm{Beta}(1,\gamma)$. We set $u_K = 1$ to truncate at level K, so that $\beta_K = 1 - \sum_{k<K}\beta_k$ and $\beta_k = 0$ for $k > K$. The source-specific weights then have a finite-dimensional Dirichlet prior on the K-simplex, and the approximation becomes exact as $K \to \infty$. We introduce latent cluster assignments $Z_i \in \{1, \ldots, K\}$ such that the latent location of observation i is $\theta^*_{Z_i} - \mu_{s_i}$. The full hierarchical model after truncation reads:

$$
\begin{aligned}
\log T_i \mid \cdot &\sim N\Big(\eta_i,\, \sigma^2_{s_i}\Big), && \text{(S26)}\\
\eta_i &= m_0^{\mathrm{sh}}(x_i) + (1 - s_i)\, d(x_i) + a_i\, \tau(x_i) + a_i(1 - s_i)\, c(x_i) + \theta^*_{Z_i} - \mu_{s_i}, && \text{(S27)}\\
m_0^{\mathrm{sh}} &\sim \mathrm{BART}(J_{\mathrm{sh}}, k_{\mathrm{sh}}, \alpha, \beta), && \text{(S28)}\\
d &\sim \mathrm{BART}(J_d, k_d, \alpha, \beta), && \text{(S29)}\\
\tau &\sim \mathrm{BART}(J_\tau, k_\tau, \alpha_\tau, \beta_\tau), && \text{(S30)}\\
c &\sim \mathrm{BART}(J_c, k_c, \alpha_c, \beta_c), && \text{(S31)}\\
Z_i \mid \boldsymbol{\pi}_{s_i} &\sim \mathrm{Categorical}(\boldsymbol{\pi}_{s_i}), && \text{(S32)}\\
\boldsymbol{\pi}_s \mid \boldsymbol{\beta},\, M_s &\sim \mathrm{Dirichlet}(M_s\beta_1, \ldots, M_s\beta_K), \qquad s \in \{0, 1\}, && \text{(S33)}\\
\boldsymbol{\beta} \mid \gamma &\sim \mathrm{Stick}_K(\gamma), && \text{(S34)}\\
\theta^*_k &\overset{\mathrm{iid}}{\sim} N(0, \sigma^2_\theta), \qquad k = 1, \ldots, K, && \text{(S35)}\\
\sigma^2_s &\sim \nu\lambda/\chi^2_\nu, \qquad s \in \{0, 1\}, && \text{(S36)}\\
\gamma &\sim \mathrm{Gamma}(a_\gamma, b_\gamma), && \text{(S37)}\\
M_s &\sim \mathrm{Gamma}(a_M, b_M), \qquad s \in \{0, 1\}, && \text{(S38)}
\end{aligned}
$$

where $\mathrm{Stick}_K(\gamma)$ is the truncated stick-breaking construction given above. We use a default $K = 50$ and track the largest occupied component index across iterations to confirm that it stays well below K.[1]

S.3.2 Outer Gibbs sampler

One iteration of the sampler consists of three blocks executed in order: updates of the four forests $\{\mathrm{sh}, d, \tau, c\}$, an update of the HDPM error block, and augmentation of the censored event times. We update the forests sequentially against partial residuals. We update the HDPM error block as a single unit because its conditional structure is internally coupled. The augmentation step at the end of the iteration updates the latent $\log T_i$ for use in the next iteration. Algorithm S1 summarises one iteration.

Algorithm S1: Outer Gibbs sampler, one iteration.

Input: Current forests $(\mathcal{T}_j^f, \mathcal{H}_j^f)$ for $f \in \{\mathrm{sh}, d, \tau, c\}$; current augmented event times $\log T_i$; HDPM state $(\boldsymbol{\theta}^*, \boldsymbol{\beta}, \boldsymbol{\pi}_0, \boldsymbol{\pi}_1, \gamma, M_0, M_1, \sigma_0^2, \sigma_1^2)$.

foreach *forest* $f \in \{\mathrm{sh}, d, \tau, c\}$ **do**
- **for** $j = 1$ **to** J_f **do**
  - Update $(\mathcal{T}_j^f, \mathcal{H}_j^f)$ by one BART backfitting step on the partial residual R_i^f, $i \in \mathcal{I}_f$;

Update the HDPM error block (Algorithm S2);

for $i = 1$ **to** n *with* $\delta_i \neq 1$ **do**
- Draw $\log T_i$ from its truncated normal full conditional;

[1] In our simulation studies the number of occupied components did not exceed 25, so the default $K = 50$ leaves substantial headroom.

S.3.3 Forest updates

We update the four forests sequentially by extended Bayesian backfitting (Hastie and Tibshirani, 2000; Chipman et al., 2010). We define a partial residual R_i^f on the active subset $\mathcal{I}_f$ of observations for each forest $f \in \{\mathrm{sh}, d, \tau, c\}$:

$$R_i^{m_0^{\mathrm{sh}}} = \log T_i - (1 - s_i)\, d(x_i) - a_i\, \tau(x_i) - a_i(1 - s_i)\, c(x_i) - (\theta^*_{Z_i} - \mu_{s_i}), \tag{S39}$$

$$R_i^d = \log T_i - m_0^{\mathrm{sh}}(x_i) - a_i\, \tau(x_i) - a_i\, c(x_i) - (\theta^*_{Z_i} - \mu_0), \tag{S40}$$

$$R_i^\tau = \log T_i - m_0^{\mathrm{sh}}(x_i) - (1 - s_i)\, d(x_i) - (1 - s_i)\, c(x_i) - (\theta^*_{Z_i} - \mu_{s_i}), \tag{S41}$$

$$R_i^c = \log T_i - m_0^{\mathrm{sh}}(x_i) - d(x_i) - \tau(x_i) - (\theta^*_{Z_i} - \mu_0), \tag{S42}$$

with active subsets

$$\mathcal{I}_{m_0^{\mathrm{sh}}} = \{1, \dots, n\}, \quad \mathcal{I}_d = \{i : s_i = 0\}, \quad \mathcal{I}_\tau = \{i : a_i = 1\}, \quad \mathcal{I}_c = \{i : a_i = 1,\, s_i = 0\}. \tag{S43}$$

Each R_i^f subtracts the contributions of the other three forests and the current cluster shift $\theta^*_{Z_i} - \mu_{s_i}$ from $\log T_i$, restricted to the subset on which f is active. Conditional on the cluster assignment Z_i and the atoms θ^*, the residual R_i^f is Gaussian with mean $f(x_i)$ and variance $\sigma^2_{s_i}$ for $i \in \mathcal{I}_f$. The BART step on forest f operates on this Gaussian working response.

Within forest f, we update each tree by a standard BART backfitting step on R_i^f. We update the tree structure $\mathcal{T}_j^f$ by a Metropolis-Hastings step with birth-death proposals (Chipman et al., 1998; Kapelner and Bleich, 2016). We then draw the leaf step heights $\mathcal{H}_j^f$ from their conjugate Gaussian posterior. The Gaussian leaf prior is conjugate to the Gaussian working response, and the posterior mean and variance at each terminal node are available in closed form.

S.3.4 HDPM error block

The HDPM error block updates the cluster assignments, atoms, top-level sticks, source-specific weights, concentration parameters, and per-source residual variances. All full conditionals are conjugate. Throughout, we write $\mid \cdot$ for conditioning on the data and on the current values of all other parameters. Algorithm S2 summarises one update of the block.

We sample the cluster assignments from

$$\mathbb{P}(Z_i = k \mid \cdot) \;\propto\; \pi_{s_i,k}\, \phi\!\left(\frac{R_i - (\theta^*_k - \mu_{s_i})}{\sigma_{s_i}}\right), \qquad k = 1, \dots, K. \tag{S44}$$

We sample the atoms in the unconstrained parameterisation and reapply the centring deterministically (Yang et al., 2010). Let $n_{sk} = \sum_i \mathbb{1}\{s_i = s, Z_i = k\}$ and $\tilde{R}_{sk} = \sum_{i: s_i = s,\, Z_i = k} (R_i + \mu_s)$. The two sources pool through the precision weights σ_s^{-2} in both the mean and the variance of the conjugate update:

$$\theta^*_k \mid \cdot \;\sim\; N\!\left(\frac{\sum_s \sigma_s^{-2}\, \tilde{R}_{sk}}{\sigma_\theta^{-2} + \sum_s \sigma_s^{-2}\, n_{sk}},\; (\sigma_\theta^{-2} + \sum_s \sigma_s^{-2}\, n_{sk})^{-1}\right), \qquad k = 1, \dots, K. \tag{S45}$$

We then recompute $\mu_s = \sum_k \pi_{sk} \theta^*_k$ and the centred atoms $\theta^*_k - \mu_s$.

We update the top-level stick weights by the auxiliary table-count augmentation of Teh et al. (2006). For each (s, k) with $n_{sk} > 0$ we sample the auxiliary table count via the Antoniak representation (Antoniak, 1974):

$$t_{sk} \;=\; \sum_{j=1}^{n_{sk}} B_{sk,j}, \qquad B_{sk,j} \;\sim\; \text{Bernoulli}\!\left(\frac{M_s\beta_k}{M_s\beta_k + j - 1}\right), \tag{S46}$$

and we set $t_{sk} = 0$ when $n_{sk} = 0$. With $t_{\cdot k} = t_{0k} + t_{1k}$, we update the truncated sticks by

$$u_k \mid \cdot \;\sim\; \text{Beta}\left(1 + t_{\cdot k},\; \gamma + \sum_{l>k} t_{\cdot l}\right), \qquad k = 1, \dots, K-1, \tag{S47}$$

with $u_K = 1$ and $\beta_k = u_k \prod_{l<k}(1 - u_l)$. The source-specific weights then follow

$$\boldsymbol{\pi}_s \mid \cdot \;\sim\; \text{Dirichlet}(M_s\beta_1 + n_{s1},\, \dots,\, M_s\beta_K + n_{sK}), \qquad s \in \{0, 1\}, \tag{S48}$$

after which we recompute μ_s and the centred atoms.

We update the concentration parameters γ and M_0, M_1 by the auxiliary-variable scheme of Escobar and West (1995). See Teh et al. (2006, Appendix A) for the derivation. Let K^* denote the number of occupied top-level components and $t_{\cdot\cdot} = \sum_{s,k} t_{sk}$ the total table count. We draw $\eta_\gamma \mid \gamma, t_{\cdot\cdot} \sim \text{Beta}(\gamma + 1, t_{\cdot\cdot})$ and then sample γ from a mixture of two Gamma distributions with mixing weight ω_γ:

$$\gamma \mid \cdot \;\sim\; \omega_\gamma\, \text{Gamma}(a_\gamma + K^*,\, b_\gamma - \log\eta_\gamma) + (1 - \omega_\gamma)\, \text{Gamma}(a_\gamma + K^* - 1,\, b_\gamma - \log\eta_\gamma), \tag{S49}$$

with $\omega_\gamma/(1 - \omega_\gamma) = (a_\gamma + K^* - 1)/[t_{\cdot\cdot}(b_\gamma - \log\eta_\gamma)]$. The analogous update for M_s replaces K^* by the source-specific table count $t_{s\cdot} = \sum_k t_{sk}$ and uses $\eta_{M,s} \mid M_s, n_s \sim \text{Beta}(M_s + 1, n_s)$.

The conjugate update of the per-source residual variance closes the block:

$$\sigma_s^2 \mid \cdot \;\sim\; \text{InverseGamma}\left(\frac{\nu + n_s}{2},\; \frac{\nu\lambda + \text{SS}_s}{2}\right), \qquad \text{SS}_s = \sum_{i:S_i = s} \big(R_i - (\theta^*_{Z_i} - \mu_s)\big)^2, \qquad s \in \{0, 1\}. \tag{S50}$$

Algorithm S2: Update of the HDPM error block.

Input: Partial residuals R_i from (S25); current atoms $\boldsymbol{\theta}^*$, sticks $\boldsymbol{\beta}$, source weights $\boldsymbol{\pi}_0, \boldsymbol{\pi}_1$, concentrations γ, M_0, M_1, scales σ_0^2, σ_1^2.

for $i = 1$ **to** n **do**
- Sample cluster assignment Z_i from the categorical full conditional (S44);

for $k = 1$ **to** K **do**
- Update the unconstrained atom θ_k^* from its conjugate Gaussian posterior (S45);

Recompute $\mu_s = \sum_k \pi_{sk}\theta_k^*$ and the centred atoms;

Sample auxiliary table counts t_{sk} via the Antoniak representation;
Update top-level sticks u_k from Beta posteriors;
Recompute $\beta_k = u_k \prod_{l<k}(1 - u_l)$;

foreach $s \in \{0, 1\}$ **do**
- Update source weights $\boldsymbol{\pi}_s$ from a Dirichlet posterior;

Recompute μ_s and the centred atoms;

Update concentrations γ, M_0, M_1 via the Escobar-West auxiliary-variable scheme;

foreach $s \in \{0, 1\}$ **do**
- Update the residual scale σ_s^2 from its inverse-gamma posterior (S50);

S.3.5 Data augmentation for censored event times

Each iteration closes with data augmentation of the unobserved event times for the censored observations (Tanner and Wong, 1987). Subjects with $\delta_i = 1$ contribute the observed $\log T_i = \log L_i = \log R_i$ and need no augmentation. Subjects with $\delta_i = 0$ have a right-censored event time and we draw

$$\log T_i \mid \cdot \;\sim\; \mathrm{TruncatedNormal}\Big(\eta_i,\, \sigma^2_{s_i};\, [\log L_i, \infty)\Big)\,, \tag{S51}$$

where η_i is the conditional mean of $\log T_i$ defined in (S27). Subjects with $\delta_i = 2$ have an interval-censored event time and we draw

$$\log T_i \mid \cdot \;\sim\; \mathrm{TruncatedNormal}\Big(\eta_i,\, \sigma^2_{s_i};\, [\log L_i,\, \log R_i]\Big)\,. \tag{S52}$$

The augmented values serve as the working response for the next iteration.

S.3.6 Hyperparameters of the nonparametric error distribution

The nonparametric error model carries three sets of hyperparameters: the per-source residual scales σ^2_s, the concentration parameters γ, M_0, M_1, and the base-measure variance σ^2_θ. We collect their priors and calibration here.

We place a scaled-inverse-χ^2 prior $\sigma^2_s \sim \nu\lambda/\chi^2_\nu$ on the per-source residual scales, with $\nu = 3$ and λ calibrated to the empirical residual variance (Chipman et al., 2010). We update each σ^2_s by the inverse-gamma conjugate step (S50).

We treat the concentration parameters γ, M_0, M_1 as unknown and estimate them. We place independent Gamma hyperpriors $\gamma \sim \mathrm{Gamma}(a_\gamma, b_\gamma)$ and $M_s \sim \mathrm{Gamma}(a_M, b_M)$, with $a_\gamma = a_M = 2$ and $b_\gamma = b_M = 0.1$. This places prior mean 20 and prior mode 10 on each concentration. We update them by the Escobar-West auxiliary-variable scheme of the HDPM error block.

The base-measure variance σ^2_θ of the unconstrained atoms is calibrated by a Yamato-type approximation (Yamato, 1984). The quantity $\sum_k \pi_{sk}(\theta^*_k - \mu_s)^2/\sigma^2_\theta$ is approximately χ^2_1. We match the induced prior on the marginal residual variance $\mathrm{Var}(\varepsilon \mid S = s)$ in tail probability to a preliminary parametric estimate $\hat{\sigma}^2_\varepsilon$. We apply the matching within each source separately, with a default tail probability $q = 0.5$.

S.3.7 BART hyperparameters and initialisation

The BART tree-prior hyperparameters $(J_f, k_f, \alpha_f, \beta_f)$ are set as in the main text. We initialise the BART forests by the standard Chipman et al. (2010) initialisation. We initialise the cluster labels by k-means on the residuals of a preliminary parametric log-normal AFT fit. We initialise the HDPM weights at $\boldsymbol{\beta} = (1/K, \ldots, 1/K)$, $\boldsymbol{\pi}_s = \boldsymbol{\beta}$, and the concentrations at $\gamma = M_0 = M_1 = 1$.

S.4 Additional results for the simulation study

We report the simulation under varying between-source heterogeneity λ_d here. Figure S1 gives the CATE metrics across the grid of λ_d. The main text discusses these results.

We give the full comparison with the deep-learning competitors here. Table S1 reports each metric across the (λ_d, λ_u) grid, with $\lambda_d, \lambda_u \in \{0, 1, 2\}$. The Bayesian fusion forest attains the lowest RMSE at every grid point.

S.4.1 Robustness to the covariate dimension

We report the efficiency of the fusion relative to the trial-only baseline as the covariate dimension p grows. Figure S2 shows the ratio of the fusion to the trial-only RMSE and posterior variance, alongside the oracle that is handed only the active covariates. Both ratios stay below one across the whole range, so the fusion improves on the trial-only analysis at every p. The RMSE ratio drifts upward as p grows, because the fusion must also learn the high-dimensional baseline, deviation, and confounding surfaces. The oracle ratio instead falls, which confirms that this erosion is the cost of searching the noise covariates rather than a failure of borrowing. The posterior-variance ratio decreases with p, so the fusion's precision advantage over the trial-only baseline in fact widens.

S.4.2 The nonparametric error distribution

We assess the source-specific hierarchical Dirichlet process mixture (HDPM) that the Bayesian fusion forest places on the residual. We isolate the residual model by setting $\lambda_u = \lambda_d = 0$, so the two sources share the same structural mean and differ only in their error distribution. We compare three residual priors within the fusion forest: a single Gaussian, a Dirichlet process mixture pooled across the two sources (one residual law for both), and the proposed source-specific HDPM (shared atoms, source-specific weights and scale). We escalate the residual difficulty over five settings. Setting 1 draws both errors from the standard normal, so the

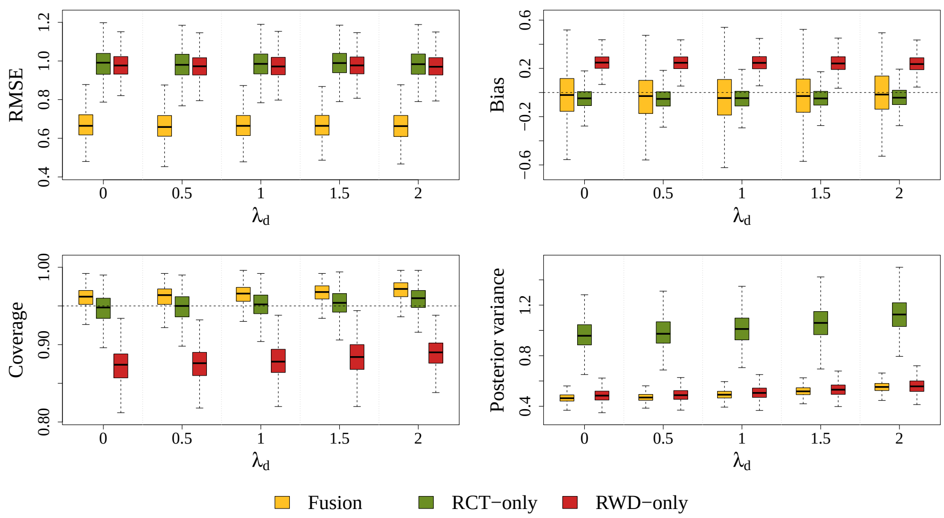


Figure S1: CATE metrics under varying between-source heterogeneity λ_d, with the unmeasured confounding held at $\lambda_u = 1$. Panels, methods, and reference lines as in Figure 1 of the main text.

λ_d	λ_u	BFF	DNN (RCT, S)	DNN (RCT, T)	DNN (pool, S)	DNN (pool, T)
RMSE						
0	0	0.68 [0.53, 0.87]	1.11 [0.98, 1.28]	1.24 [0.89, 1.75]	1.08 [0.94, 1.27]	1.11 [0.70, 3.04]
1	0	0.68 [0.53, 0.85]	1.11 [0.98, 1.27]	1.25 [0.88, 1.74]	1.08 [0.93, 1.27]	1.11 [0.73, 2.94]
2	0	0.67 [0.52, 0.85]	1.12 [0.98, 1.29]	1.27 [0.89, 1.84]	1.09 [0.95, 1.27]	1.22 [0.83, 3.06]
0	1	0.68 [0.54, 0.85]	1.11 [0.97, 1.29]	1.23 [0.91, 1.74]	1.10 [0.96, 1.27]	1.09 [0.71, 3.12]
1	1	0.68 [0.52, 0.86]	1.12 [0.98, 1.26]	1.25 [0.89, 1.77]	1.10 [0.96, 1.28]	1.13 [0.74, 3.02]
2	1	0.68 [0.53, 0.84]	1.13 [0.98, 1.28]	1.26 [0.88, 1.86]	1.12 [0.96, 1.28]	1.23 [0.83, 3.13]
0	2	0.68 [0.53, 0.86]	1.11 [0.97, 1.26]	1.25 [0.91, 1.72]	1.16 [1.02, 1.33]	1.12 [0.72, 3.16]
1	2	0.69 [0.54, 0.87]	1.12 [0.99, 1.27]	1.25 [0.90, 1.76]	1.17 [1.01, 1.33]	1.19 [0.78, 3.26]
2	2	0.68 [0.53, 0.87]	1.12 [0.98, 1.29]	1.24 [0.87, 1.85]	1.17 [1.01, 1.33]	1.25 [0.82, 3.20]
Bias						
0	0	−0.01 [−0.42, 0.38]	−0.17 [−0.55, 0.23]	−0.21 [−0.70, 0.33]	−0.13 [−0.39, 0.14]	−0.02 [−0.47, 0.74]
1	0	−0.03 [−0.44, 0.36]	−0.20 [−0.53, 0.19]	−0.23 [−0.70, 0.27]	−0.13 [−0.42, 0.14]	0.01 [−0.48, 0.76]
2	0	−0.04 [−0.46, 0.36]	−0.20 [−0.53, 0.20]	−0.28 [−0.81, 0.28]	−0.12 [−0.40, 0.18]	−0.04 [−0.58, 0.88]
0	1	−0.03 [−0.42, 0.37]	−0.18 [−0.52, 0.20]	−0.23 [−0.72, 0.29]	0.11 [−0.14, 0.39]	0.09 [−0.36, 0.92]
1	1	−0.03 [−0.45, 0.37]	−0.19 [−0.54, 0.21]	−0.25 [−0.77, 0.26]	0.11 [−0.16, 0.38]	0.10 [−0.39, 0.91]
2	1	−0.01 [−0.42, 0.41]	−0.20 [−0.54, 0.18]	−0.25 [−0.80, 0.32]	0.10 [−0.17, 0.38]	0.06 [−0.51, 0.99]
0	2	−0.03 [−0.43, 0.37]	−0.20 [−0.55, 0.16]	−0.25 [−0.75, 0.26]	0.36 [−0.03, 0.65]	0.19 [−0.30, 1.08]
1	2	−0.02 [−0.43, 0.39]	−0.19 [−0.53, 0.21]	−0.24 [−0.76, 0.31]	0.35 [−0.01, 0.65]	0.19 [−0.35, 1.25]
2	2	−0.00 [−0.43, 0.42]	−0.19 [−0.53, 0.23]	−0.24 [−0.77, 0.33]	0.32 [−0.05, 0.65]	0.21 [−0.43, 1.25]
Coverage						
0	0	0.96 [0.92, 0.98]	0.38 [0.28, 0.50]	0.91 [0.76, 0.99]	0.41 [0.31, 0.52]	0.99 [0.97, 1.00]
1	0	0.96 [0.93, 0.98]	0.37 [0.27, 0.48]	0.91 [0.78, 0.99]	0.42 [0.32, 0.54]	0.99 [0.97, 1.00]
2	0	0.97 [0.94, 0.99]	0.37 [0.27, 0.48]	0.92 [0.79, 0.99]	0.43 [0.34, 0.54]	0.99 [0.97, 1.00]
0	1	0.96 [0.93, 0.98]	0.38 [0.27, 0.48]	0.91 [0.77, 0.99]	0.44 [0.33, 0.56]	0.99 [0.96, 1.00]
1	1	0.96 [0.93, 0.99]	0.38 [0.28, 0.49]	0.91 [0.78, 0.99]	0.44 [0.34, 0.56]	0.99 [0.97, 1.00]
2	1	0.97 [0.94, 0.99]	0.37 [0.27, 0.48]	0.92 [0.79, 0.99]	0.45 [0.34, 0.56]	0.99 [0.96, 1.00]
0	2	0.96 [0.93, 0.99]	0.37 [0.27, 0.49]	0.91 [0.77, 0.99]	0.45 [0.36, 0.56]	0.99 [0.96, 1.00]
1	2	0.96 [0.93, 0.99]	0.37 [0.27, 0.48]	0.91 [0.79, 0.99]	0.45 [0.35, 0.58]	0.99 [0.96, 1.00]
2	2	0.97 [0.94, 0.99]	0.37 [0.27, 0.49]	0.93 [0.78, 0.99]	0.45 [0.35, 0.57]	0.99 [0.96, 1.00]
Posterior variance						
0	0	0.48 [0.41, 0.56]	0.07 [0.04, 0.12]	1.15 [0.74, 1.94]	0.12 [0.05, 0.38]	3.96 [0.97, 11.76]
1	0	0.50 [0.44, 0.58]	0.07 [0.04, 0.14]	1.18 [0.74, 2.06]	0.12 [0.05, 0.34]	4.19 [1.05, 11.48]
2	0	0.56 [0.49, 0.65]	0.07 [0.04, 0.15]	1.35 [0.74, 2.98]	0.13 [0.05, 0.42]	4.72 [1.18, 13.42]
0	1	0.49 [0.42, 0.57]	0.07 [0.04, 0.11]	1.10 [0.71, 1.79]	0.12 [0.05, 0.40]	3.72 [1.02, 10.24]
1	1	0.52 [0.44, 0.60]	0.07 [0.04, 0.11]	1.13 [0.74, 1.93]	0.13 [0.05, 0.42]	4.13 [1.09, 11.12]
2	1	0.58 [0.50, 0.67]	0.07 [0.04, 0.12]	1.21 [0.76, 2.35]	0.13 [0.05, 0.45]	5.19 [1.24, 14.93]
0	2	0.52 [0.45, 0.61]	0.07 [0.04, 0.10]	1.08 [0.72, 1.65]	0.15 [0.06, 0.45]	3.60 [1.08, 10.63]
1	2	0.55 [0.46, 0.63]	0.06 [0.04, 0.10]	1.09 [0.74, 1.62]	0.15 [0.06, 0.44]	4.06 [1.12, 11.81]
2	2	0.61 [0.52, 0.71]	0.07 [0.04, 0.11]	1.12 [0.75, 1.77]	0.14 [0.07, 0.45]	5.27 [1.33, 15.58]

Table S1: CATE metrics for the Bayesian fusion forest (BFF) and the four deepAFT competitors across the (λ_d, λ_u) grid. Each entry is the mean over simulation replicates, with the 2.5–97.5% range in brackets. R and P denote the RCT-only and naive-pool fits; S and T denote the S-learner and T-learner.

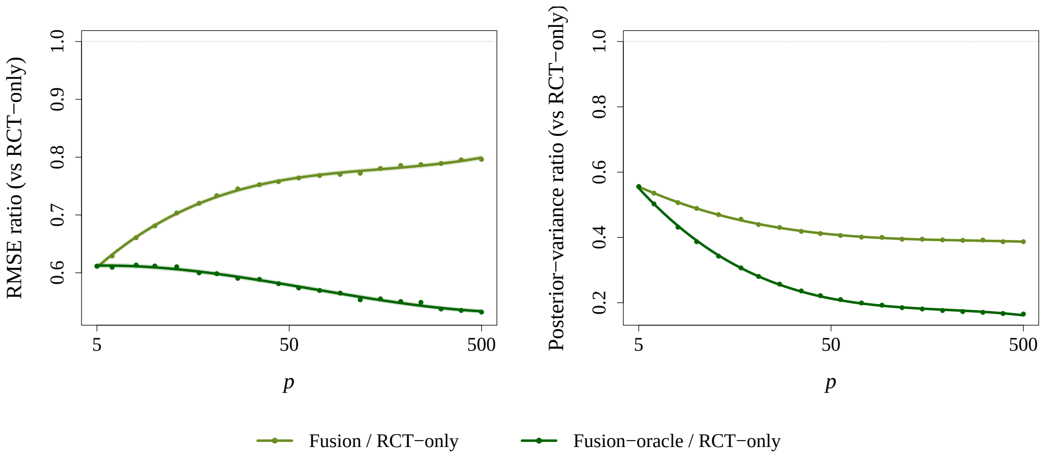


Figure S2: Efficiency of the Bayesian fusion forest relative to the trial-only baseline as the covariate dimension p grows, at $\lambda_d = \lambda_u = 1$. Left: ratio of RMSE; right: ratio of posterior variance. A ratio below one favours the fusion. The fusion oracle is the fusion forest given only the active covariates. Shaded bands are 95% Monte Carlo intervals.

Gaussian prior is correctly specified. Settings 2–5 give the real-world source a markedly larger scale than the trial (a standard-deviation ratio of about three), which the single-scale Gaussian and pooled priors cannot represent, and add increasing non-Gaussianity: a pure scale gap (2), a skewed real-world error (3), a bimodal real-world error (4), and multimodal errors in both sources with shared component locations but different weights (5). To make the residual model bite, we censor heavily and coarsely: the trial is right-censored at circa 50% and the real-world data are interval-censored at quartile visits, so many observations enter the likelihood only through the error distribution over wide intervals. We report two targets. The first is the CATE $\tau(x)$, our primary estimand. The second is the full conditional-mean outcome $\mathbb{E}[\log T \mid X, A, S]$, the sum of the four forests, which the residual model affects through every component rather than through the treatment contrast alone.

Table S2 reports the CATE metrics over the combined population. At the correctly-specified null the three priors are indistinguishable, so the flexible residual models cost nothing when a Gaussian would do. From Setting 2 onward the HDPM attains the lowest RMSE at every setting, together with the lowest, or tied-lowest, bias. The Gaussian prior is the most biased throughout, and its bias grows as the error departs from normality, because under heavy coarse censoring a misspecified residual distribution biases the imputation of the censored events. The pooled mixture removes much of this bias by modelling the shape flexibly, but, tied to a single residual law and scale for both sources, it cannot match the source-specific HDPM once the sources differ. Coverage is near nominal, and slightly conservative, for all three priors, and posterior variance is comparable, so the accuracy gains of the HDPM do not come at a calibration cost. The pattern is the same in the trial- and real-world-only populations.

Residual prior	RMSE	Bias	Coverage	Post. var.
Setting 1 — Gaussian, identical (null)				
Gaussian	0.745	−0.099	0.956	0.557
Shared DP	0.744	**−0.064**	0.957	0.561
Source HDPM	**0.743**	−0.066	0.959	0.576
Setting 2 — Gaussian, per-source scale gap				
Gaussian	0.751	−0.146	0.964	0.627
Shared DP	0.745	−0.111	0.964	0.629
Source HDPM	**0.739**	**−0.099**	0.965	0.634
Setting 3 — Skewed RWD, scale gap				
Gaussian	0.757	−0.179	0.961	0.613
Shared DP	0.745	**−0.119**	0.962	0.608
Source HDPM	**0.740**	−0.121	0.963	0.613
Setting 4 — Bimodal RWD, scale gap				
Gaussian	0.758	−0.138	0.965	0.655
Shared DP	0.754	−0.117	0.966	0.666
Source HDPM	**0.742**	**−0.096**	0.967	0.661
Setting 5 — Multimodal, shared atoms, weight + scale gap				
Gaussian	0.771	−0.150	0.966	0.691
Shared DP	0.768	−0.136	0.967	0.706
Source HDPM	**0.752**	**−0.107**	0.968	0.694

Table S2: CATE metrics by residual prior across the five error settings, over the combined population and averaged over simulation replicates. “Source HDPM” is the proposed source-specific hierarchical Dirichlet process mixture; “Shared DP” pools one Dirichlet process mixture across sources. The best RMSE and bias per setting are in bold. Coverage is the 95% credible-interval coverage; “Post. var.” is the mean posterior variance of $\hat{\tau}(x)$.

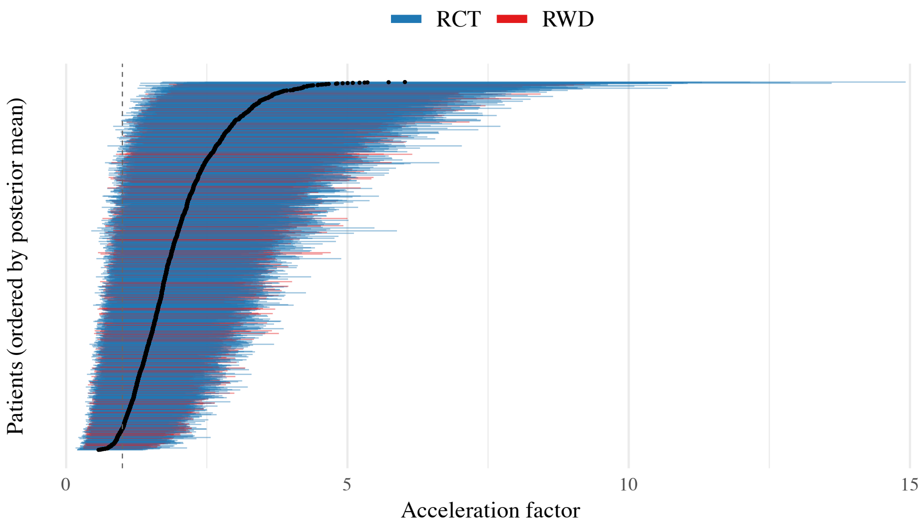


Figure S3: Subject-level acceleration factor for every patient in the trial-aligned cohort, from the trial-only analysis, ordered by posterior mean. Vertical bars are pointwise 95% credible intervals; the dashed line at 1 marks no effect. The intervals are wider than those of the fusion forest in Figure 5 of the main text.

S.5 Additional results for the data analysis

We collect here the cohort description, the trial-only comparison, and the setup of the deep-learning competitor. Table S3 characterises the trial-aligned analysis cohort. Figure S3 gives the subject-level acceleration factor from the trial-only analysis, the comparator for the fusion estimates in the main text. Table S4 compares the precision of the subject-level estimates between the trial-only and fusion analyses.

	ACTG 175 (RCT)	MACS (RWD)
Age, y	34 [29, 41]	39 [34, 44]
CD4, cells/mm^3	337 [263, 422]	263 [178, 365]
CD8, cells/mm^3	918 [670, 1246]	849 [659, 1153]
Calendar year	1992	1991 [1991, 1992]
Prior ART, y	0.4 [0, 2.1]	1.0 [0, 1]
Non-white, n (%)	404 (23)	78 (21)

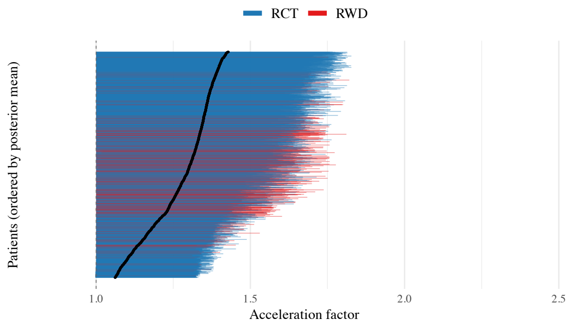
Table S3: Characteristics of the trial-aligned analysis cohort of 2144 patients, by data source. Continuous variables are reported as median [IQR], categorical variables as n (%).

S.5.1 Comparison with a deep-learning competitor

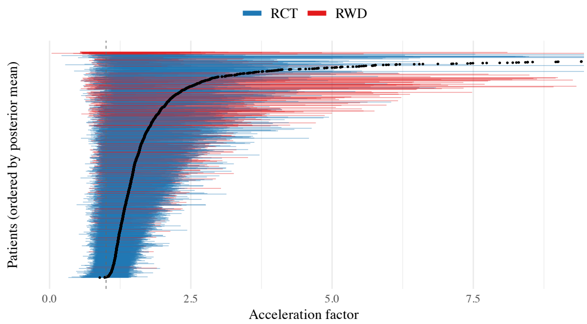
We benchmark the fusion against a deep accelerated failure time network (Norman et al., 2024), the same black-box competitor used in the simulation study. We fit the network as an S-learner and a T-learner (Künzel et al., 2019), on the trial alone and on a naive pool of both sources with a source indicator. The architecture matches the simulation: two hidden layers of eight and six units with rectified-linear activations and a linear output. We tune the learning rate, the momentum, and the weight decay by ten-fold cross-validation on the concordance index. We obtain intervals from 100 stratified bootstrap resamples, since the network has no posterior.

	Trial-only	Fusion	Reduction
Average 95% CrI width	3.16	1.21	62%
Average posterior variance	0.808	0.098	88%

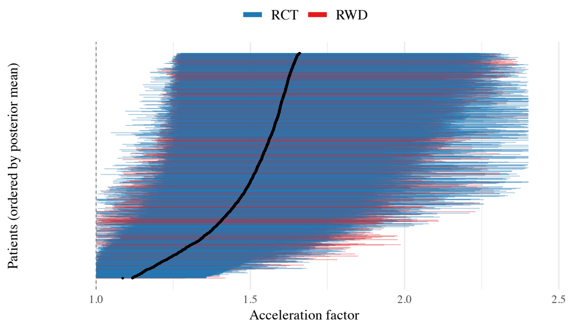
Table S4: Precision of the subject-level acceleration factor over the combined cohort, under the trial-only and fusion analyses. Both quantities are averaged across patients on the acceleration-factor scale.



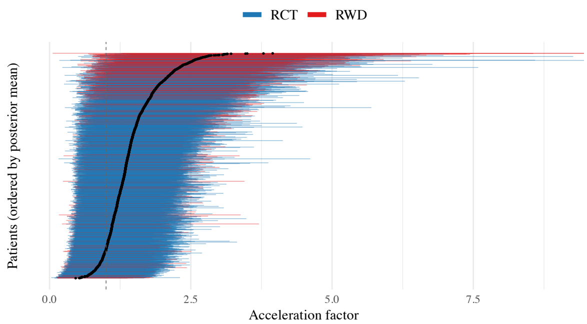
(a) Trial-only, S-learner



(b) Trial-only, T-learner



(c) Pooled, S-learner



(d) Pooled, T-learner

Figure S4: Subject-level acceleration factor from the deep-learning competitor over the combined RCT + MACS cohort, by learner (S, T) and training design (trial-only, pooled), ordered by the bootstrap mean. Bars are the 95% bootstrap interval; colour indicates the data source. The estimates and their intervals vary markedly across configurations, in contrast to the single calibrated fusion caterpillar (Figure 5) of the main text.

The network requires right-censored data. We therefore approximate the interval-censored MACS deaths by their right-censored last-observation encoding, the same encoding used by the trial-only forest. This approximation discards the interval information and is, if anything, favourable to the competitor. The main text reports the headline comparison. Figure S4 shows the subject-level acceleration factor from each network.

The two learners in Figure S4 fail in opposite ways. The S-learner uses a single network with the treatment as one input among the covariates. It learns a near-additive treatment effect, so the acceleration factor is nearly constant and exceeds one for every patient (Künzel et al., 2019). The estimates therefore lie in a narrow band above one. This orderly picture reflects regularisation toward a homogeneous effect rather than calibrated certainty, and the simulation shows that the S-learner intervals undercover. The T-learner instead fits a separate network per arm and differences the two fits. These fits are noisy on the smaller per-arm samples, so the estimates range from well below one to above three. Neither competitor recovers the calibrated subject-level effect of the fusion.

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